

Introduction to the Course

Applied Parallel Programming



KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN

fit@hcmus

Introduction

- ☐ This course is the continuation of “Parallel Programming” course
- ☐ If you have not taken “Parallel Programming” course, you will need to study by yourself materials of “Parallel Programming” course
- ☐ This course has (**mostly**) **one project**

Giải mã hiện tượng Nvidia: 8 tháng có 1.000 tỉ USD nhờ chip AI



▶ Nghe đọc bài 6:04 | 1x |  ▾

Tuổi Trẻ trên **Google News**

news google

Mất hơn hai thập niên để đạt giá trị 1.000 tỉ USD nhưng chỉ tám tháng để có được 1.000 tỉ USD tiếp theo, Nvidia đang tạo ra một hiện tượng trên sàn chứng khoán Mỹ nhờ vào việc tiên phong trong chip AI.

THÔNG TIN VÀ TRUYỀN THÔNG > KINH TẾ SỐ

Thứ bảy, 24/02/2024 - 13:52

Nvidia phá kỷ lục chứng khoán Mỹ, tăng vốn hoá 277 tỉ USD chỉ trong một ngày



Thế Vinh

→ [Xem các bài viết của tác giả](#)



Zalo



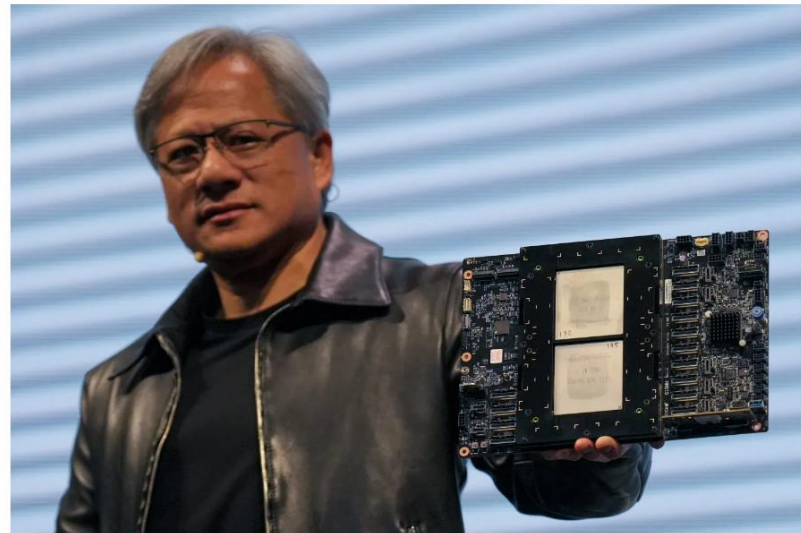
Theo dõi VietNamNet trên **Google News**

“Cỗ máy in tiền” Nvidia tiếp tục cho thấy sức hút lớn đối với nhà đầu tư sau khi công ty công bố kết quả kinh doanh vượt dự báo.

Nvidia shatters stock market record by adding over \$230 billion in value in one day. Here's why it's dominating the AI chip race

BY DYLAN SLOAN

February 23, 2024 at 2:32 AM GMT+7

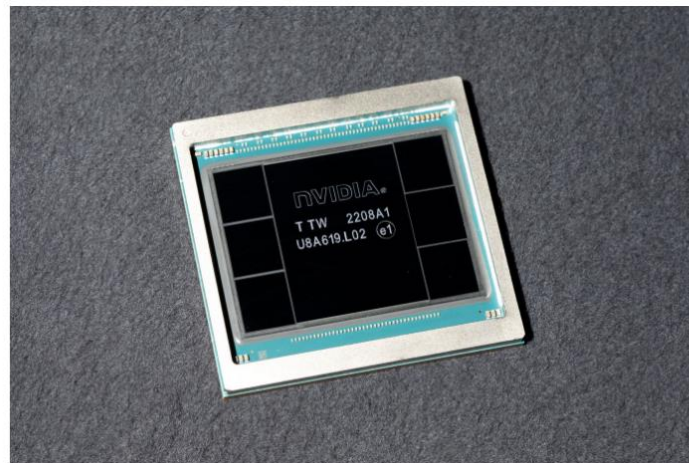


Nvidia CEO Jensen Huang (holding one of the chip designer's Hopper units) has led the company to the forefront of the AI chip race.

WALID BERRAZEG—SOPA IMAGES/LIGHTROCKET/GETTY IMAGES

The AI Chip Behind Nvidia's Supersonic Stock Rally: QuickTake

- Nvidia's technology has few rivals at the task of training AI
- The chip's successors are likely to be the H200, then the B100



What will you do in the project?

- ☐ It's similar to the final project in “Parallel Programming” course
- ☐ You will:
 - ☐ Choose an application (and a algorithm behind it) which is interesting but needs to speed up
 - ☐ Understand the algorithm behind and implement it sequentially
 - ☐ Speed up this sequential program step by step: first parallelizing, then optimizing further (like “Parallel Programming” course, here we focus on parallel programming on GPU)

What will you do in the project?

- ☐ The application (and the algorithm behind it) in the project of this course is expected to be **more time-consuming** to implement and hopefully more interesting
- ☐ You will have more time to do
- ☐ You will receive feedback from friends and Teacher many times during the process of doing the project

What will you do in the project?

- You will use CUDA Python (Numba) which:
 - ▣ Help to code faster than CUDA C/C++
- But still keep the beauty of parallel programming as CUDA C/C++: let programmers think and design how to divide work among threads as well as how to coordinate these threads so that they will not interfere with each other
- There are Python libs such as CuPy, CuDF, cuML, ... which just magically run parallel on GPU and don't require users to think about how to parallelize

What will you do in the project?

- ☐ The project will be done in teams, ~2 students / team

What will you do in the project?

Final products of the project:

- ☐ Runnable report file (.ipynb file) describing the **application**, the **algorithm** behind it, and the **process of parallelizing/optimizing** to speed up the sequential version I expect that this file is written clearly
- ☐ Team plan file and Github repo's commit history
 - ☐ I expect that the workload is quite balanced between team members and everyone understands the work of each other

Suggested applications for the project

1. Training a Machine Learning model (e.g. XGBoost)

Input: a training data set and necessary hyper-parameter values

Output: a trained model

Training a Machine Learning model is slow; we all know this

Suggested applications for the project

2. Changing image background

Input: a portrait image with bad-looking background and a good-looking background image

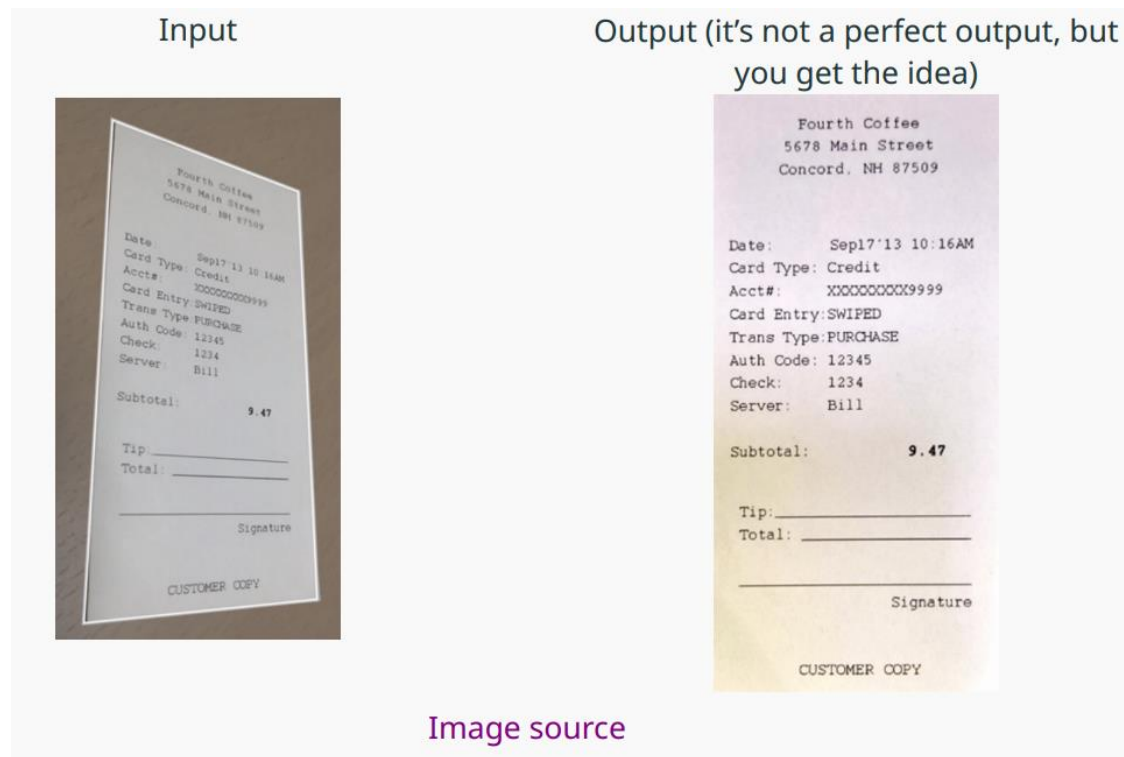
Output: the portrait image with good-looking background

If we can process an image fast enough, we can process in real-time (like in Zoom meeting)

Suggested applications for the project

3. Enhancing document image

If we can process an image fast enough, we can process in **real-time**



Suggested applications for the project

- ☐ Random Forest
- ☐ HOG (histogram of oriented gradient) + SVM for image classification
- ☐ Parallel Chess Engine
- ☐ JPEG: Image compression
- ☐ Customers Segmentation based on their Credit Card usage behaviour
- ☐ Image classification using Convolutional Neural Network (CNN)
- ☐ Concurrent linear hashing
- ☐ Thay đổi nền ảnh bằng phương pháp Poisson matting
- ☐ Parallel LSTM (a recurrent neural network) training for sequence prediction from sequential data
- ☐ Song song hóa mô hình XGBoost để nhận dạng loại bệnh thực vật
- ☐ Parallelize KMeans - credit card segmentation
- ☐ Parallel Collaborative Filtering - Recommender System

Course plan

- ☐ Week 01-05: preparation for the project
 - ☐ Mindset
 - ☐ Tools: Git & Github, Google Colab, Python, Numpy, Numba
 - ☐ The process of parallelizing/optimizing a program to make it run faster
 - ☐ **Teamwork** (to make sure the workload is quite balanced between team members and everyone understand the work of each other), **writing/speaking** (about your project), **feedback** (about your friends' projects)
- ☐ Week 06-13: weekly project presentation and feedback
- ☐ Week 14 or 15: official project presentation

How is your final grade computed?

- **Final grade** = 0.3 **Participation grade** + 0.7 **Project grade**
- **Participation grade**: Grade for participation in weekly project presentation and feedback
 - ▣ Participation is your **personal business**; its grade can be different for members in your team
- **Project grade**: Grade for official project presentation at the end of the course
 - ▣ Project is your **team's business**; its grade will be the same for members in your team and $\approx$ average of individual member grades in the official project presentation

Demo

- ☐ Parallelizing matrix multiplication to make it run faster